space available to it, the electron, like a claustrophobic human, starts to shake uncontrollably. It flies about its tiny cell at high speed, kicking with great force against adjacent electrons in their cells. This *degenerate motion,* as physicists call it, cannot be stopped by cooling the matter. Nothing can stop it; it is forced on the electron by the laws of quantum mechanics, even when the matter is at absolute zero temperature.

This degenerate motion is a consequence of a feature of matter that Newtonian physicists never dreamed of, a feature called *wave/particle duality:* Every kind of particle, according to quantum mechanics, sometimes behaves like a wave, and every kind of wave sometimes behaves like a particle. Thus, waves and particles are really the same thing, a "thing" that sometimes behaves like a wave and sometimes like a particle; see Box 4.1.

Electron degeneracy is easily understood in terms of wave/particle duality. When matter is compressed to high densities, and each electron inside the matter gets confined to an extremely small cell squeezed up against neighboring electrons' cells, the electron begins to behave in part like a wave. The wavelength of the electron wave (the distance between its crests) cannot be larger than the electron's cell; if it were, the wave would extend beyond the cell. Now, particles with very short wavelengths are necessarily highly energetic. (A common example is the particle associated with an electromagnetic wave, the photon. An X-ray photon has a wavelength far shorter than that of a photon of light, and as a result X-ray photons are far more energetic than photons of light. Their higher energies enable the X-ray photons to penetrate human flesh and bone.)

In the case of an electron inside very dense matter, the electron's short wavelength and accompanying high energy imply rapid motion, and this means that the electron must fly around inside its cell, behaving like an erratic, high-speed mutant: half particle, half wave. Physicists say that the electron is "degenerate," and they call the pressure that its erratic high speed motion produces "electron degeneracy pressure." There is no way to get rid of this degeneracy pressure; it is an inevitable consequence of confining the electron to such a small cell. Moreover, the higher the matter's density, the smaller the cell, the shorter the electron wavelength, the higher the electron energy, the faster the electron's motion, and thus the larger its degeneracy pressure. In ordinary matter with ordinary densities,